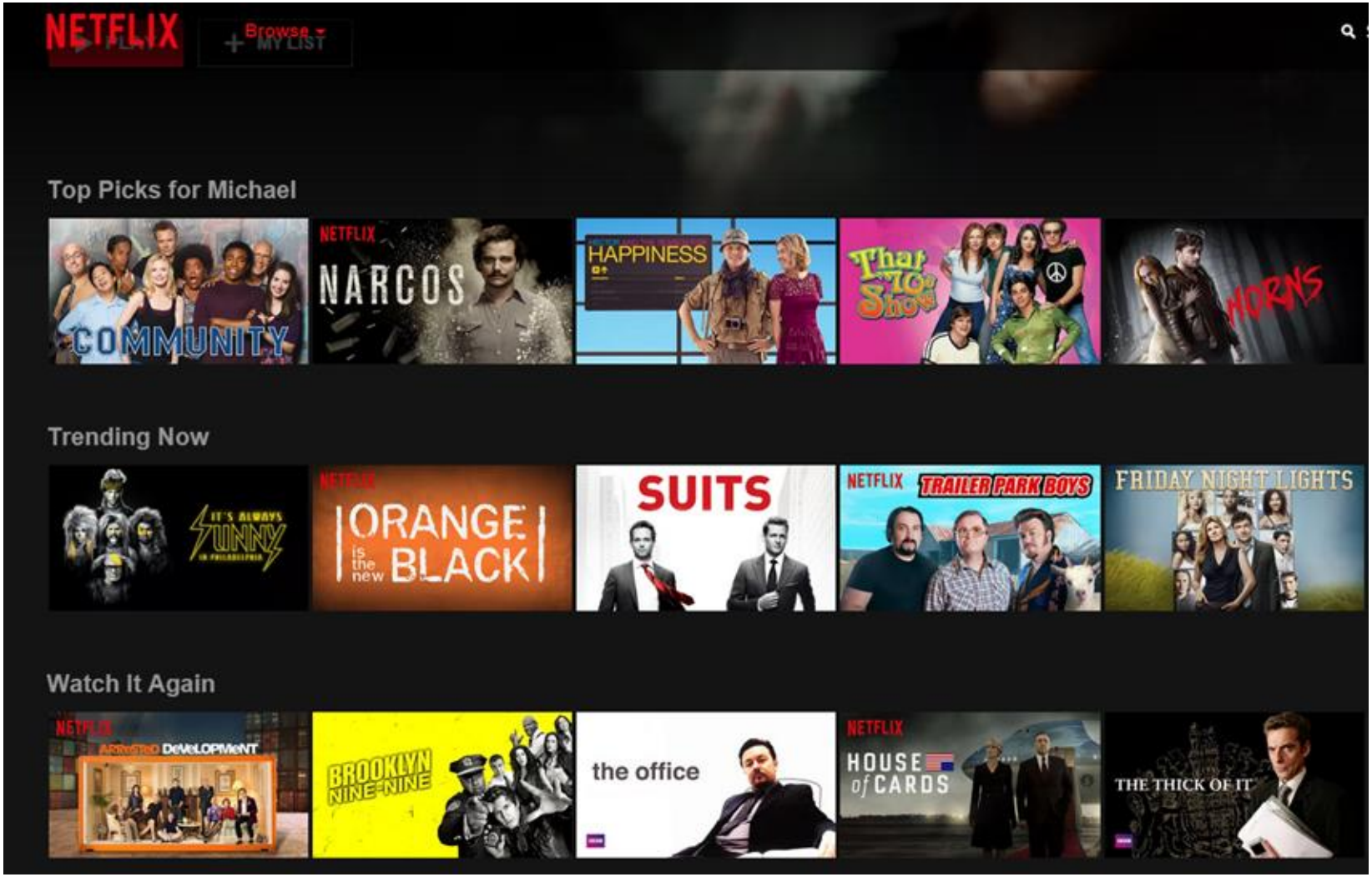
Recommender Systems



Agenda

- **Introduction**
- User-based collaborative filtering
- Content-based filtering
- Historical examples

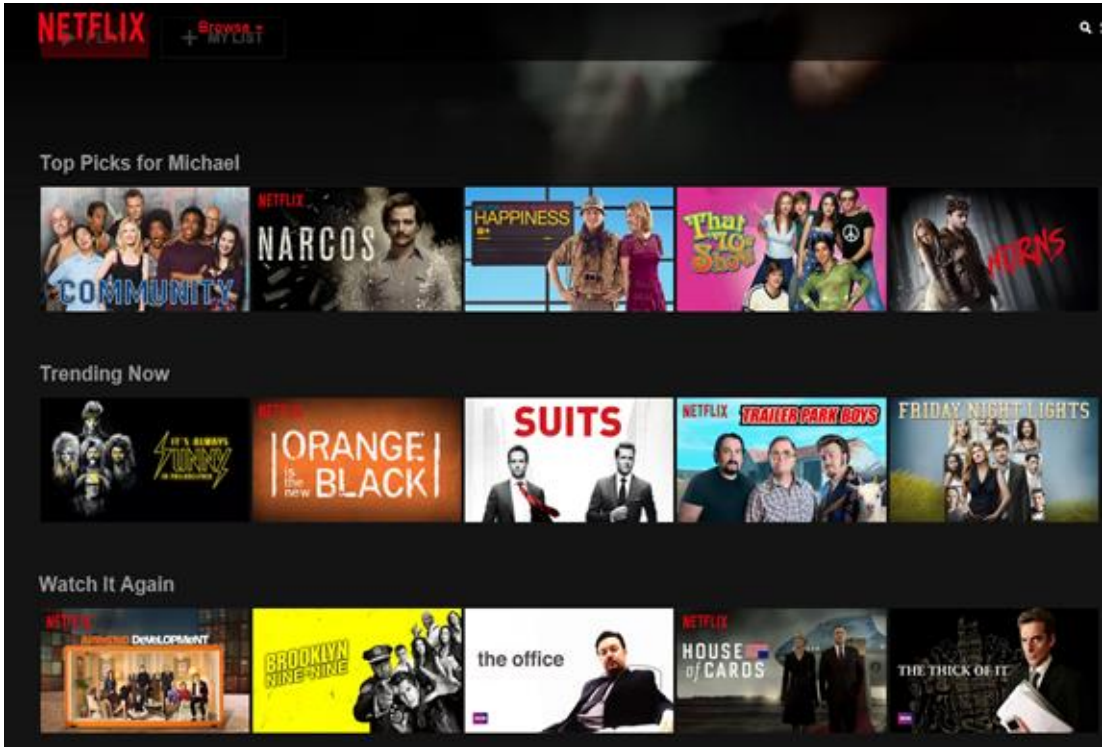
Introduction

- *Recommender Systems* (e.g., Amazon, YouTube, Home Depot, etc.) predict what you'll like

- Put another way: they predict your preferences
- What are some examples?

Example: Movies, shows

- Netflix, Amazon, Hulu, et al.

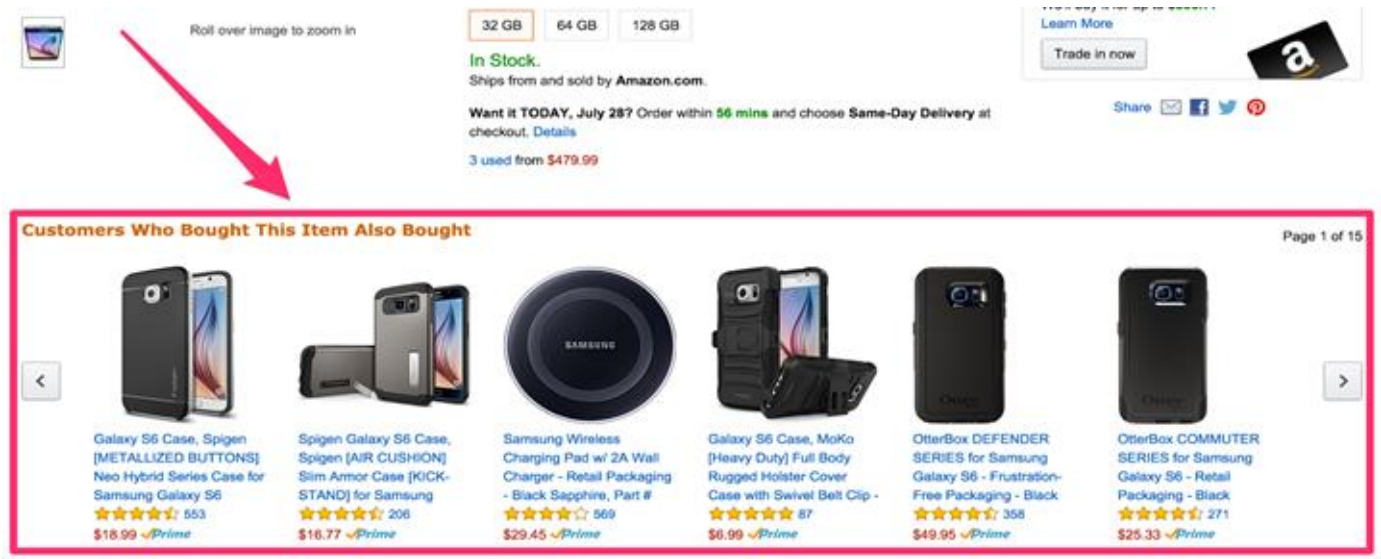


Example: Music

- Spotify, Apple Music, et al.

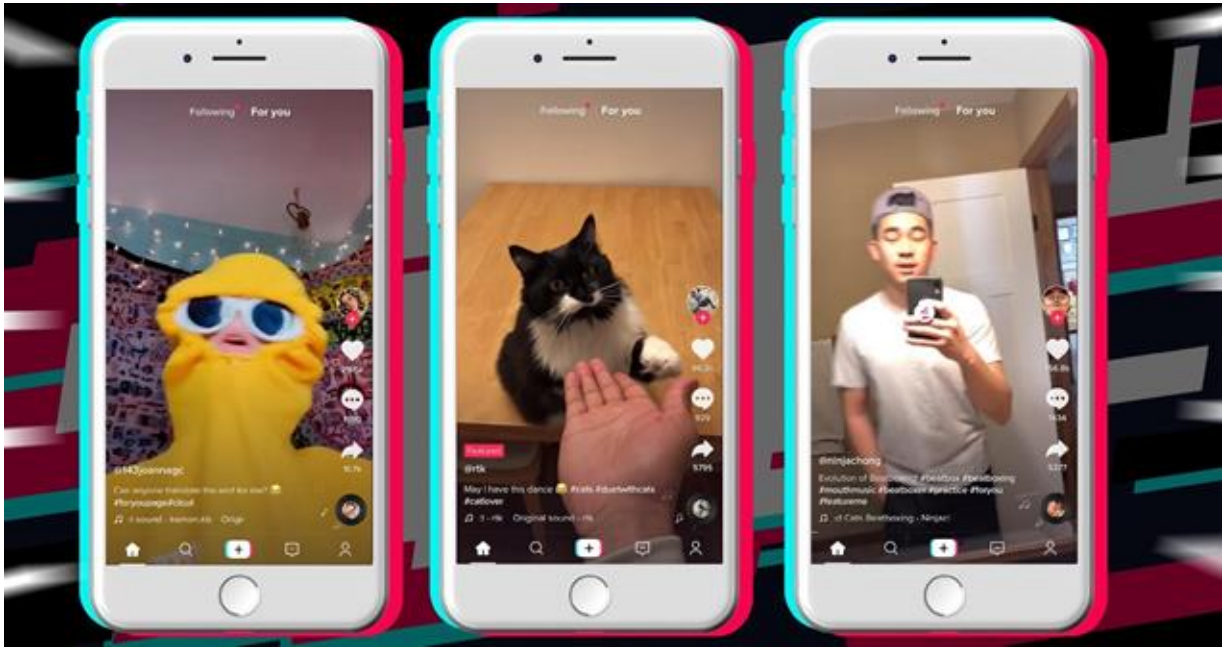
Example: Products

- Amazon, Home Depot, et al.



Example: Videos

- TikTok, YouTube, et al.



Recommendation challenges

- Millions of users
- Millions of items
- Sparse data (most ratings missing)
- Must run in real-time

Data collection

- Explicit data collection
  - Ask users to rate movies, products, whatever
  - Ask users for demographic information
- Implicit data collection
  - Web logs of past user activity
  - Timing information, e.g., how long you paused on a post before scrolling
  - Accounts you follow
  - Language and country settings

Agenda

- Introduction
- **User-based collaborative filtering**
- Content-based filtering
- Historical examples

Collaborative filtering

- Problem: How to recommend movies without knowing anything about movies?
- Solution: Recommend items liked by similar users.

- People who agreed in the past will agree in the future

Averaging

- Filling missing ratings using average of all users.

Averaging example

	W.	Panda	Youngblood	Zorro
Alice	4	2	4	4
Bob	?	2	5	1
Chris	4	2	4	?
Donna	3	?	5	1

Averaging example

	W.	Panda	Youngblood	Zorro
Alice	4	2	4	4
Bob	3.66	2	5	1
Chris	4	2	4	2
Donna	3	2	5	1

Averaging problem

- Problem: Averaging ignores uniqueness of a user
  - Terrible when there is large variation in interest
  - Movies, music are a few examples

- Solution: Nearest neighbor algorithm

Nearest neighbor algorithm

- Find another user with similar rating patterns
  - Movie ratings in this example
- Use other user's rating to "fill in the blank"
- People who agreed in the past will agree in the future

Nearest neighbor example

- Who are the nearest neighbors?

	W.	Panda	Youngblood	Zorro
Alice	4	2	4	4
Bob	?	2	5	1
Chris	4	2	4	?
Donna	3	?	5	1

Nearest neighbor example

- Bob's nearest neighbor is Donna

	W.	Panda	Youngblood	Zorro
Alice	4	2	4	4
Bob	?	2	**5**	**1**
Chris	4	2	4	?
Donna	3	?	**5**	**1**

Nearest neighbor example

- Use Donna's rating to fill in Bob's

	W.	Panda	Youngblood	Zorro
Alice	4	2	4	4
Bob	**3**	2	**5**	**1**
Chris	4	2	4	?
Donna	**3**	?	**5**	**1**

Nearest neighbor example

- Chris's nearest neighbor is Alice

	W.	Panda	Youngblood	Zorro
Alice	**4**	**2**	**4**	**4**
Bob	3	2	5	1
Chris	**4**	**2**	**4**	**4**
Donna	3	?	5	1

Nearest neighbor example

- Donna's nearest neighbor is Bob

	W.	Panda	Youngblood	Zorro
Alice	4	2	4	4
Bob	3	**2**	**5**	**1**
Chris	4	2	4	4
Donna	3	**2**	**5**	**1**

Distance metric

- How to measure distance between neighbors?
- Many possibilities, including
  - Euclidean - distance
  - Cosine – angle similarity
  - Pearson correlation coefficient
- A lot in common with document vector model

K-nearest neighbors (k-NN)

- Can we do better than selecting just one nearest neighbor?
- Select several.

1. Find the k similar users instead of 1
2. Select the most frequent score (mode)

k-NN problems

1. Find the k similar users instead of 1
2. Select the most frequent score (mode)

- How would a really popular film affect recommendations?

k-NN problems

1. Find the k closest users
2. Select the most frequent score

- How would a really popular film affect recommendations?

- The popular film will be recommended most of the time because it will often be the most frequent score
- The popular choice might not always be the best choice for a user

k-NN problems

- How would a "cult classic" film affect recommendations?
  - Enjoyed by a few, who REALLY like it
- What could you do about this?

k-NN problems

- How would a "cult classic" film affect recommendations?
  - Enjoyed by few, but they REALLY like it
- Rare film less likely to be predicted by k-NN

- Can use *inverse-user-frequency*
  - Similar to TFxIDF's inverse document frequency
  - Rarely seen movies are more influential

User-based collaborative filtering weaknesses

- Cold start
  - Need user data before you can start making accurate recommendations
- Scalability
  - Millions of users
  - Large amount of computation power
- Sparsity
  - Most users will only have rated a small subset of the overall database
  - Even the most popular items have very few ratings

Agenda

- Introduction
- User-based collaborative filtering
- **Content-based filtering**
- Historical examples

Motivation

- Problems with user-based collaborative filtering
  - Cold start, scalability, sparsity
- Solution: focus on content instead of users
  - Avoid nearest-neighbor operations on users

Content-based filtering

- *Content-based filtering:* Recommend items similar to items the user has liked in the past

- People will like the same things in the future that they liked in the past

Content-based filtering synonyms

- Content-based filtering
- AKA Content-based collaborative filtering
- AKA Item-based collaborative filtering
- AKA "Users who bought x also bought y"

Content-based filtering

- How can we compute item similarity?
  - Similar techniques to user-based collaborative filtering
  - Euclidean distance, cosine similarity, correlation, tf-idf

- User-based filtering finds similar rows (users)
- Content-based filtering finds similar columns (items)

Content-based filtering example

- W. is most similar to Youngblood
- Use likes W.? Recommend Youngblood

	W.	Panda	**Youngblood**	Zorro
Alice	**4**	2	**4**	4
Bob	**3**	2	**5**	1
Chris	**4**	2	**4**	4
Donna	**3**	2	**5**	1

Content-based filtering weaknesses

- Problem: Website doesn't know all your preferences
  - You've rated horror movies on Netflix, but you also like comedies
- Problem: Lost opportunity to introduce you to new content
- Solution: Hybrid filtering

Hybrid filtering

- *Hybrid filtering:* Combine user-based and content-based filtering
- Netflix Example:
  - Recommendations based on similar users; AND
  - Recommendations based on shows user has rated highly

Agenda

- Introduction
- User-based collaborative filtering
- Content-based filtering
- **Historical examples**

Target and pregnancy prediction

1. Predict pregnancy
2. Recommend pregnancy related products
3. Father angry when teen daughter receives ads
4. Father finds out daughter is pregnant

How Target Figured Out A Teen Girl Was Pregnant Before Her Father Did

Kashmir Hill Forbes Staff

Welcome to The Not-So Private Parts where technology & privacy collide

Every time you go shopping, you share intimate details about your consumption patterns with retailers. And many of those retailers are studying those details to figure out what you like, what you need, and which coupons are most likely to make you happy. Target TGT +0%, for example, has figured out how to data-mine its way into your womb, to figure out whether you have a baby on the way long before you need to start buying diapers.

TARGET

Target has got you in its aim

Target and pregnancy prediction

- Features were products
  - Unscented lotion, calcium, magnesium and zinc supplements
- Sent ads through the mail
- Story from 2012

Netflix prize

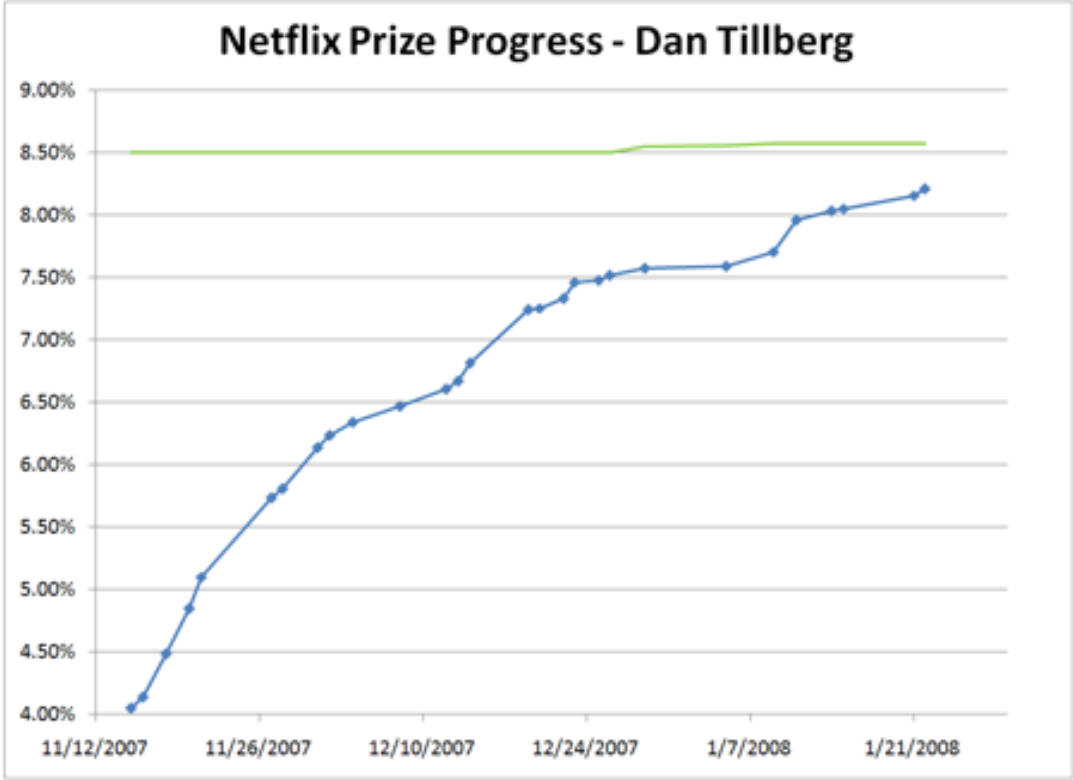
- Announced October 2, 2006
- $1M to whoever could improve NetFlix's own recommender by 10%
- Data from Netflix in the form:
  - <user, movie, date, grade>
  - Where grade is 1...5 (today it's just thumbs up/down)
  - That's it
- Training data: 100M examples from ~500k users on ~18k movies
- Ideas for how you might do this?

Netflix prize

- How did they do it?
- Among many ideas:
  - Use IMDB for director, genre, etc.
  - Some movies' grades change when reviewers grade them in a clump, suggesting a long time has passed since the movie was viewed
  - User grades depend on day of week

Netflix prize

- Dan Tillberg's progress…
- Dropped out in 2008 while in 5th place



Netflix prize

- Won on September 21, 2009

Summary

- User-based collaborative filtering: recommend items enjoyed by *users who are similar to you*
- Content-based filtering: recommend items similar to *items you have enjoyed in the past*
- Hybrid filtering: use both user- and content-based filtering at the same time